

Remote Photoplethysmography and Motion Tracking Convolutional Neural Network with Bidirectional Long Short-Term Memory: Non-Invasive Fatigue Detection Method Based on Multi-Modal Fusion

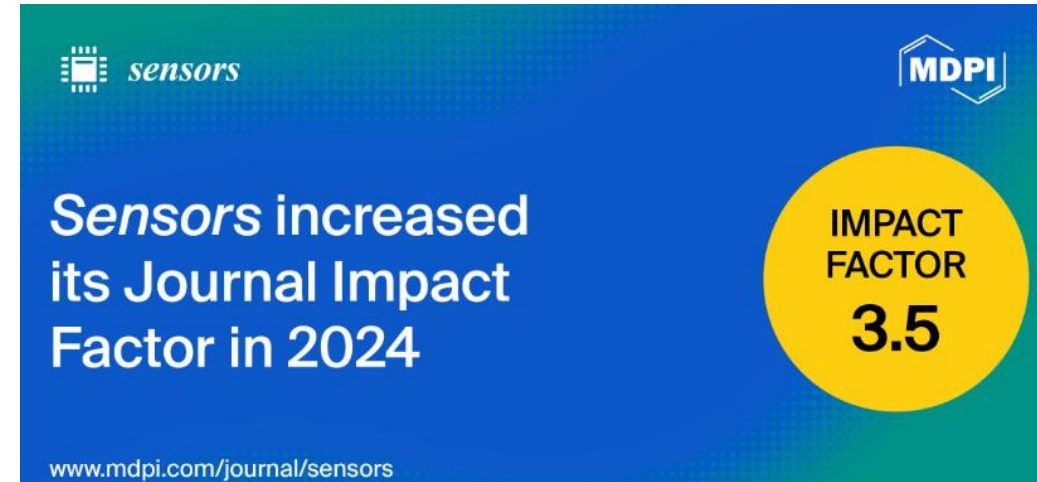
GE1.02 Shang-Wei Chang

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About the article

- Sensors received an Impact factor of 3.5 in 2024
- The authors are from Yangtze University¹, Central South University^{2,3}.



Article

Remote Photoplethysmography and Motion Tracking Convolutional Neural Network with Bidirectional Long Short-Term Memory: Non-Invasive Fatigue Detection Method Based on Multi-Modal Fusion

Lingjian Kong¹, Kai Xie^{1,*}, Kaixuan Niu¹, Jianbiao He² and Wei Zhang³

Outline

- Introduction
- System Architecture
- Results
- Conclusion
- Limitations & Future Work

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Introduction-research motivation

- Why is this important?
 - Driver fatigue, drowsiness, and sleep deprivation are primary causes of major traffic accidents.
 - Fatigue-related incidents account for 25% to 30% of all road accidents.
 - These accidents result in significant personal injury, property loss, and societal instability.
- Goal: A reliable, real-time system that can detect driver fatigue and provide timely alerts is critical for upholding traffic safety.

Introduction-Current Detection Methods & Their Limitations

- How is fatigue currently detected?
 1. Contact-Based (Physiological):
 - I. Electroencephalogram (EEG) or Electrocardiogram (ECG).
 - Pro: Highly accurate.
 - Con: Invasive. Requires wearable sensors, which are inconvenient, expensive, and impractical for daily driving.
 2. Vision-Based (Visual Features):
 - I. Monitoring eyelid movement (PERCLOS), yawning frequency, or head posture using RGB cameras.
 - Pro: Non-invasive.
 - Con: Unreliable. Highly susceptible to changing light conditions (day vs. night), shadows, and driver motion.

The Research Gap & Proposed Solution

- **The Gaps:**
 - Low Accuracy: Single-feature methods (e.g., just eyes) are not robust enough.
 - Poor Robustness: RGB cameras fail under variable lighting.
 - Ignored Temporal Data: Fatigue is a dynamic process (a time-series), not a single-moment event. Most models ignore this.
- **Proposed Solution: RPPMT-CNN-BiLSTM**
- A non-invasive, multi-modal method that addresses all three gaps:
 - Uses an Infrared (IR) Camera to solve lighting issues.
 - Fuses two modalities:
 - Physiological: Heart Rate (via remote PPG).
 - Visual: Facial Features (PERCLOS).
 - Uses a BiLSTM network to model fatigue as a time-series.

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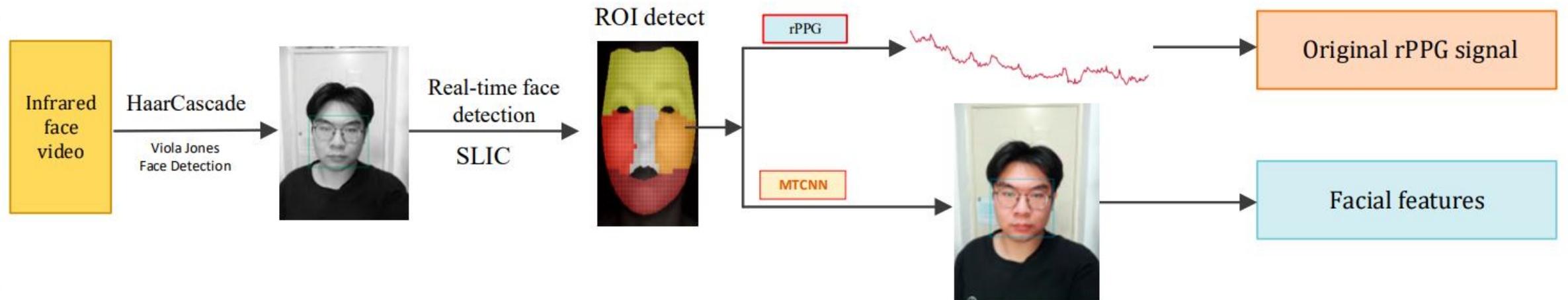
System Architecture (Overview)

A. Contactless Multi-modal Feature Extraction

B. Feature Extraction Enhancement Network

C. Multi-modal Fusion Fatigue Detection Model

A.Contactless multimodal feature extraction

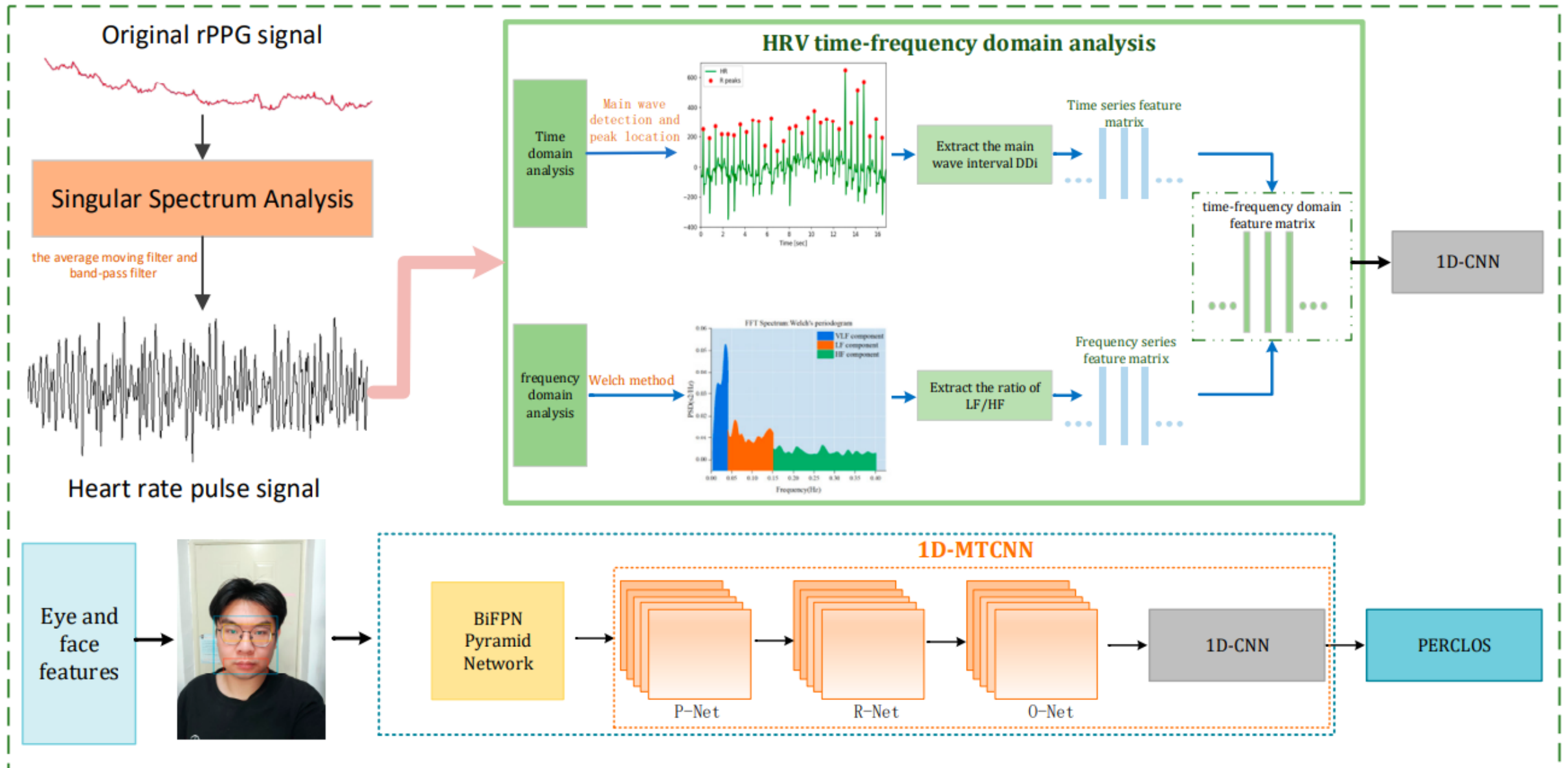


1. Haar-Cascade face detector to identify faces in all frames of the video stream
2. SLIC algorithm for superpixel skin segmentation on the detected face areas

A: Contactless Feature Extraction

- Input Source:
 - Infrared (IR) Camera: Chosen specifically to reduce errors caused by day/night lighting changes.
- **Extracted Modalities:**
- Physiological Signal (rPPG):
 - Remote Photoplethysmography (rPPG) is a non-contact way to measure blood flow.
 - The system tracks a Region of Interest (ROI) on the face (cheeks, forehead).
 - It averages pixel intensity in the IR channel to get a "heart rate pulse signal".
- Visual Signal (Facial Features):
 - Uses MTCNN (a cascaded convolutional network) for face detection and key point localization (eyes, nose, mouth).
 - Enhanced with a BiFPN pyramid network to better capture multi-scale features and improve robustness.

B.RPPMT-CNN feature extraction enhancement network



B: Feature Enhancement (Physiological)

- **Goal:** Turn the noisy, raw rPPG signal into stable metrics.
- **Signal Cleaning:**
 - Singular Spectrum Analysis (SSA): Used to separate the target heart rate signal from noise.
 - Filtering: A moving average filter and a Hamming window bandpass filter (0.8-4 Hz) are applied to remove low-frequency (e.g., breathing) and high-frequency noise.
- **Heart Rate Variability (HRV) Analysis:**
 - Uses an improved Pan-Tompkins algorithm to detect the main wave peaks (like R-peaks in an ECG).
 - Calculates the time between peaks (DD interval).
- **Feature Extraction:**
 - **Time-Domain:** Standard Deviation (SD) of DD intervals. (Fatigue increases this).
 - **Frequency-Domain:** LF/HF ratio. (Fatigue also increases this).

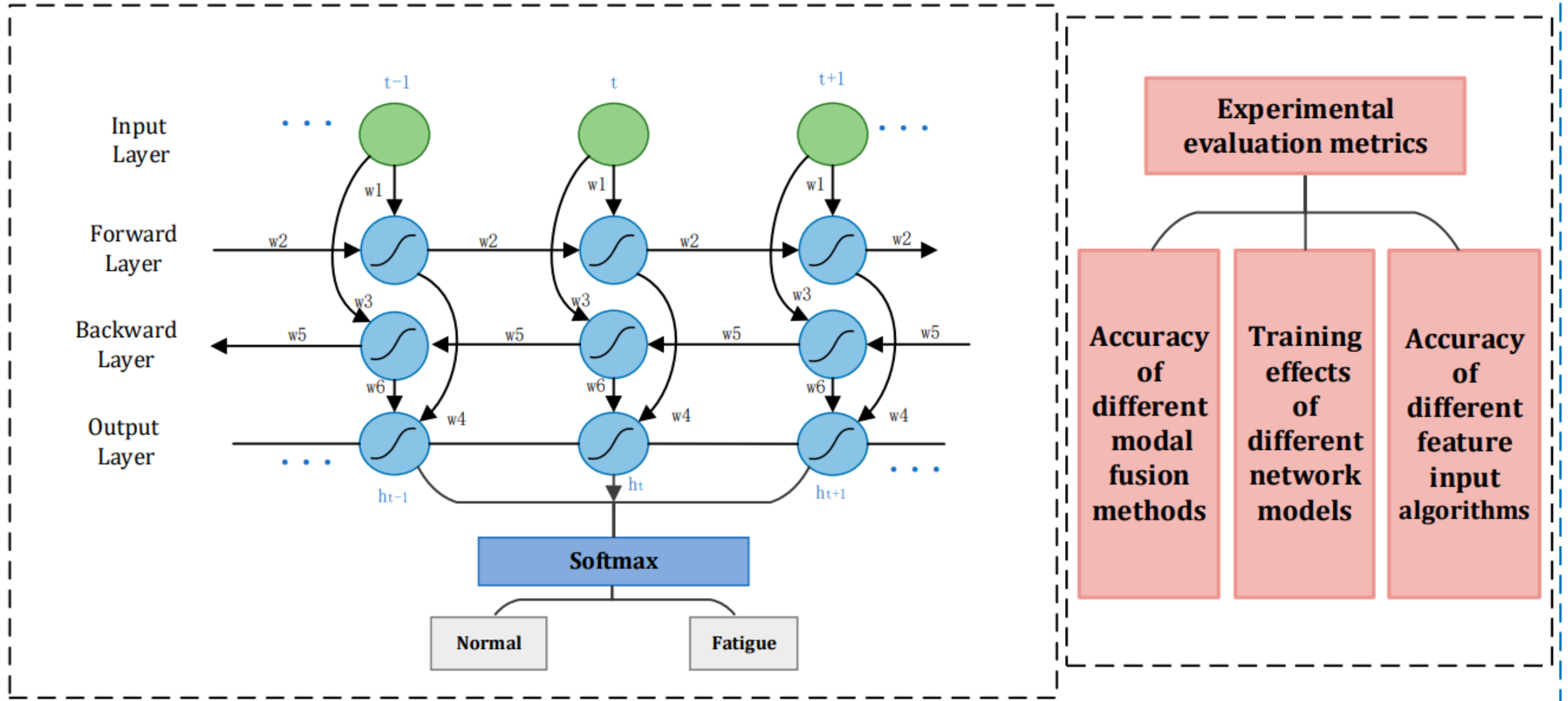
B: Feature Enhancement (Visual)

- **Goal:** Turn facial key points into a fatigue metric.
 - Algorithm: 1D-MTCNN
 - The MTCNN key point coordinates are used to precisely extract images of the eye regions.
- Metric Extraction: PERCLOS
 - The 1D-CNN model classifies each frame as "eyes open" or "eyes closed"³⁷.
 - PERCLOS (Percentage of Eye Closure) is calculated over a set time period (e.g., 60 seconds).

$$P = \frac{\text{Eyes closed frames}}{\text{The total number of frames in the detection period}} \times 100\%$$

- This value is a well-established indicator of drowsiness.

C. Multimodal fusion fatigue detection model



C: Multi-Modal Fusion Model

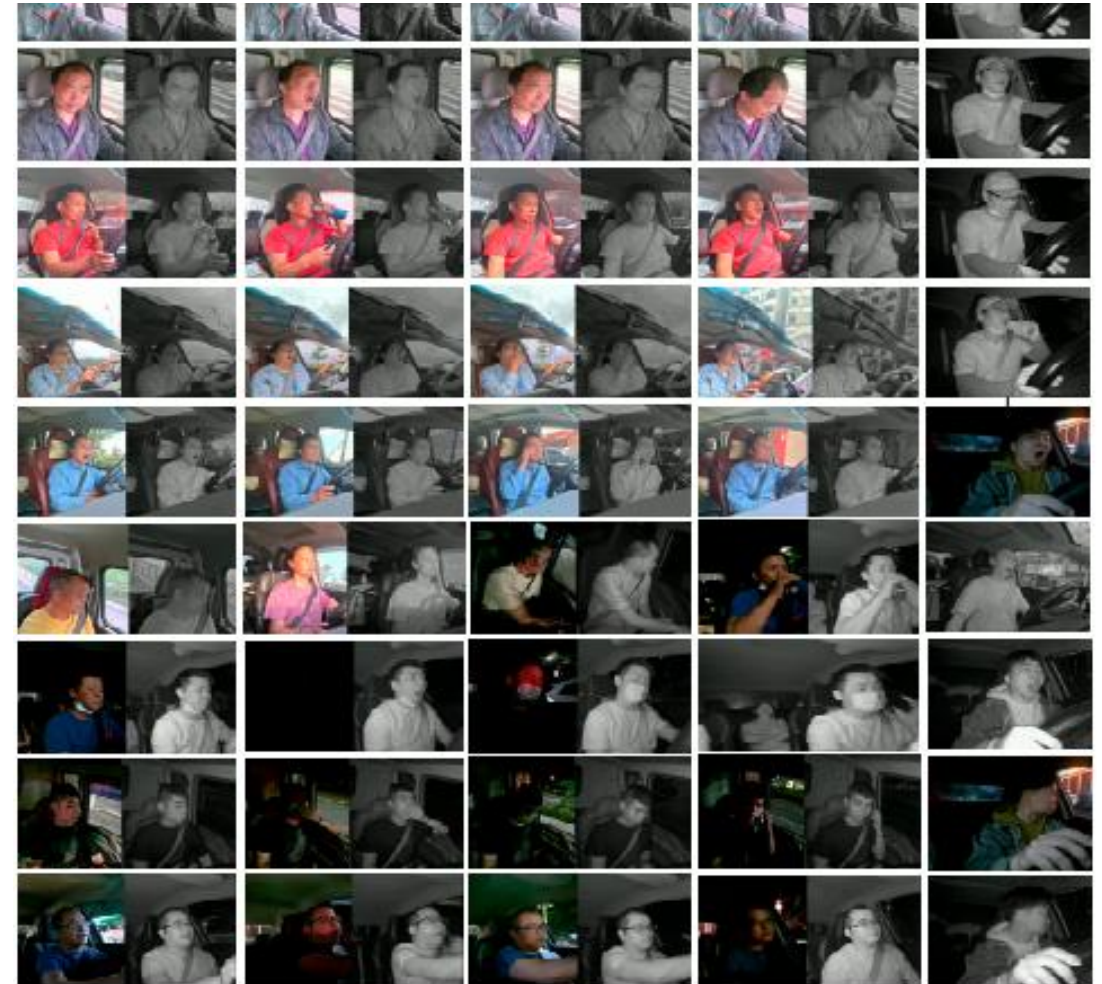
- How to combine the two features?
- **Individual Feature Modeling:**
 - The two enhanced feature sets (HRV metrics and PERCLOS values) are not immediately combined.
 - First, they are independently modeled using two separate 1D-Convolutional Neural Networks (1D-CNNs).
 - This allows each network to become an expert at finding fatigue patterns within *its own* modality (physiological vs. visual).
- **Temporal Fusion:**
 - The outputs (trained data results) from the two 1D-CNNs are then used as input for the final fusion model.
- **Final Classifier: Bidirectional LSTM (BiLSTM)**
 - This is the core of the fusion. The BiLSTM receives the feature streams from both CNNs and makes the final decision.

Why BiLSTM? (The Temporal Dimension)

- **Fatigue is a Time-Series Problem**
 - A driver's state is dynamic; it changes over time. A single snapshot is not enough.
- **LSTM (Standard):**
 - Processes data in a forward pass.
 - Uses past information to predict the present.
 - *Limitation:* It has no knowledge of what happens *after* the current moment.
- **BiLSTM (Bidirectional):**
 - Processes data in two directions:
 - 1. A forward pass (using past information).
 - 2. A backward pass (using future information).
 - This allows the model to use both past and future context to make a more accurate judgment about the current state.
 - This is ideal for fatigue, where a trend (e.g., eyes *starting* to close more) is more informative than a single event.

Experimental Setup: The MDAD Dataset

- No publicly available dataset exists with *both* RGB and Infrared multi-modal data for fatigue. So, the authors created their own.
- MDAD (Multi-Modal Driver Alertness Dataset)
 - Participants: 52 drivers (28 men, 24 women), ages 22-53.
 - Camera: Oni S500 binocular (RGB + IR).
 - Scenarios: Real driving, different vehicles, daytime and nighttime.
 - Dataset Size: 4000 60-second video clips.
 - Labels: Clips were labeled "awake" or "fatigued" using the Karolinska Sleepiness Scale (KSS).



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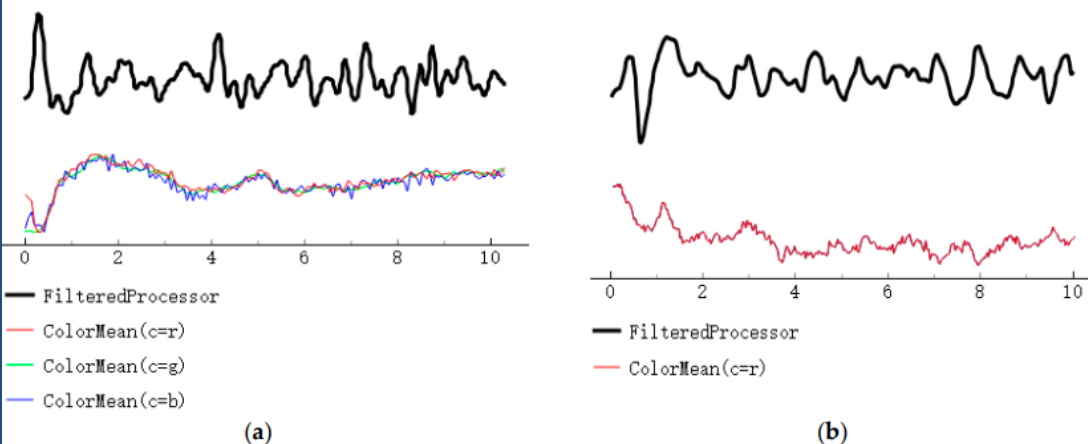
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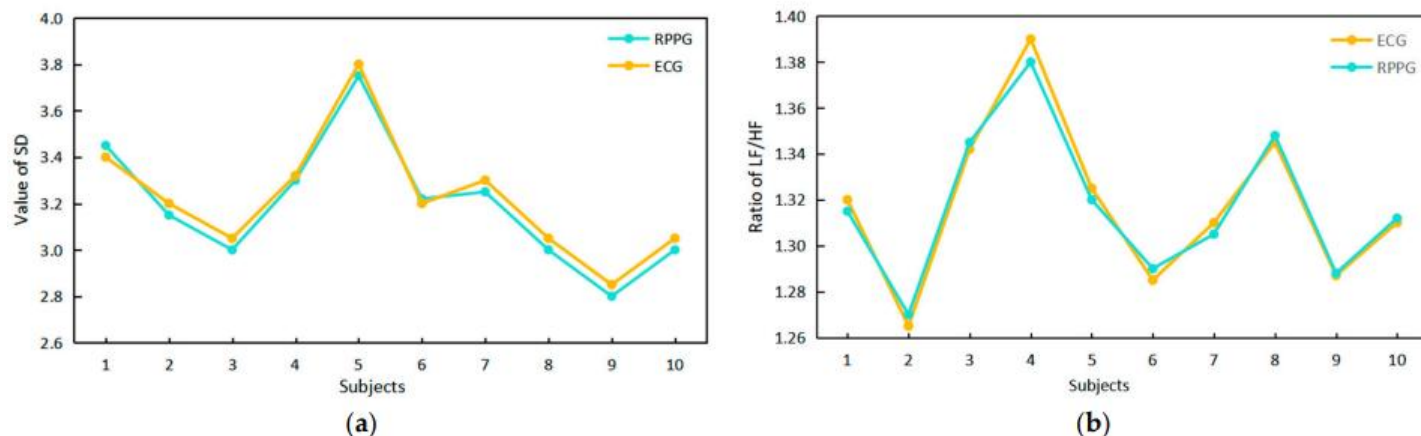
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Result 1: Is Infrared rPPG a Reliable Proxy for ECG?

1. Signal Quality: IR rPPG (b) is much cleaner and less noisy than standard RGB rPPG (a)

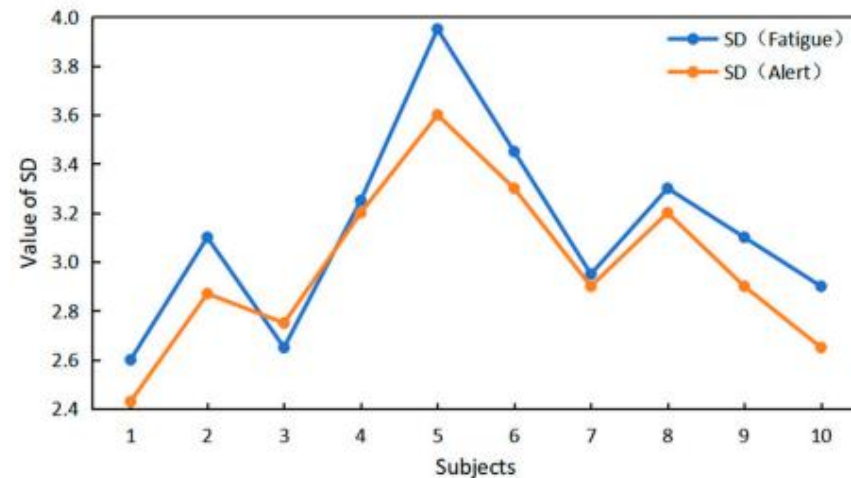


2. Metric Accuracy: The HRV metrics (SD value and LF/HF ratio) from IR rPPG (blue line) almost perfectly match the metrics from the contact ECG (yellow line).

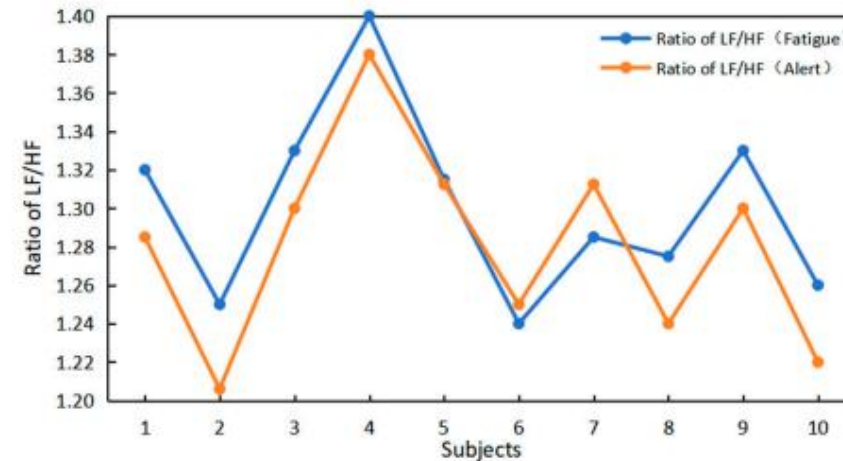


Result 2: Do These Metrics Actually Indicate Fatigue?

- Do the SD and LF/HF values change when a person is tired?



(a)



(b)

- For all subjects, both the SD value (a) and the LF/HF ratio (b) were significantly higher in the 'Fatigue' state (blue line) compared to the 'Alert' state (orange line).
- These specific HRV metrics are strong physiological indicators of fatigue.

Result 3: Which algorithms is best?

Category	Feature	Index	SVM	Random Forest	DBN	CNN	Ours
Physiological characteristics	ECG	HR, HRV	0.792	0.827	0.853	0.874	0.894
	rPPG	HRV	0.778	0.817	0.827	0.864	0.895
Visual characteristics	Eyes	PERCLOS	0.785	0.822	0.849	0.868	0.877
	Mouth	Yawn frequency	0.757	0.783	0.806	0.855	0.895
Mixed visual features	Eyes + Mouth	PERCLOS, yawn frequency	0.814	0.836	0.869	0.896	0.948
Mixing multi-class features	Eyes + Mouth + rPPG	PERCLOS, yawn frequency, HRV	0.774	0.842	0.871	0.949	0.972

Note: Bold is the best result.

SVM: Support Vector Machine (classification task)
DBN: deep belief network (feature detection)

Final Result: Comparison to other models

- How does this model stack up against other recent methods?
- The RPPMT-CNN-BiLSTM model was tested against 9 other mainstream fatigue detection algorithms on the same challenging MDAD dataset.

Method	Acc (%)
MCSVM [36]	87.3%
RF-DCM [37]	94.6%
Drowsiness detection system [28]	85%
Fatigue-Radio [33]	81.4%
FFD system [31]	97.8%
MTCNN + InceptionV3 [10]	91.1%
HDDD [38]	88.9%
CNN + DF-LSTM [39]	88.9%
Fusion system [32]	97.3%
Ours	98.2%

Note: Bold is the best result.

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Conclusion

- **A Novel Non-Invasive Method:** Proposed the **RPPMT-CNN-BiLSTM** model, which is both highly accurate (98.2%) and robust.
- **Solved Environmental Challenges:** Use of an **Infrared Camera** with rPPG and MTCNN effectively mitigates errors from day/night lighting changes.
- **Advanced Feature Enhancement:** An improved **Pan-Tompkins** algorithm successfully extracts reliable HRV from non-contact rPPG signals.
- **Superior Fusion Model:** Showed that **Multi-Modal Fusion** (Physiological + Visual) is more robust than any single modality, and that **BiLSTM** is the optimal network to capture the temporal nature of fatigue.
- **New Public Resource:** Created and validated the **MDAD dataset**, a new, challenging benchmark for future research.

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Limitations & Future Work

- **Identified Limitations / Future Directions:**
- **Computational Cost:** The BiLSTM and multi-CNN architecture is complex. Real-time implementation on low-power, in-vehicle hardware would be the next challenge.
- **Dataset Diversity:** While strong, the MDAD dataset was collected in one region. Further testing on a more globally diverse driver population (e.g., different ethnicities, skin tones) would be needed to ensure universal rPPG robustness.
- **Other Fatigue Indicators:** The model could be expanded to include other modalities, such as steering wheel movement, lane-keeping, or even audio (yawn detection).

Thank you for listening

- Q&A session